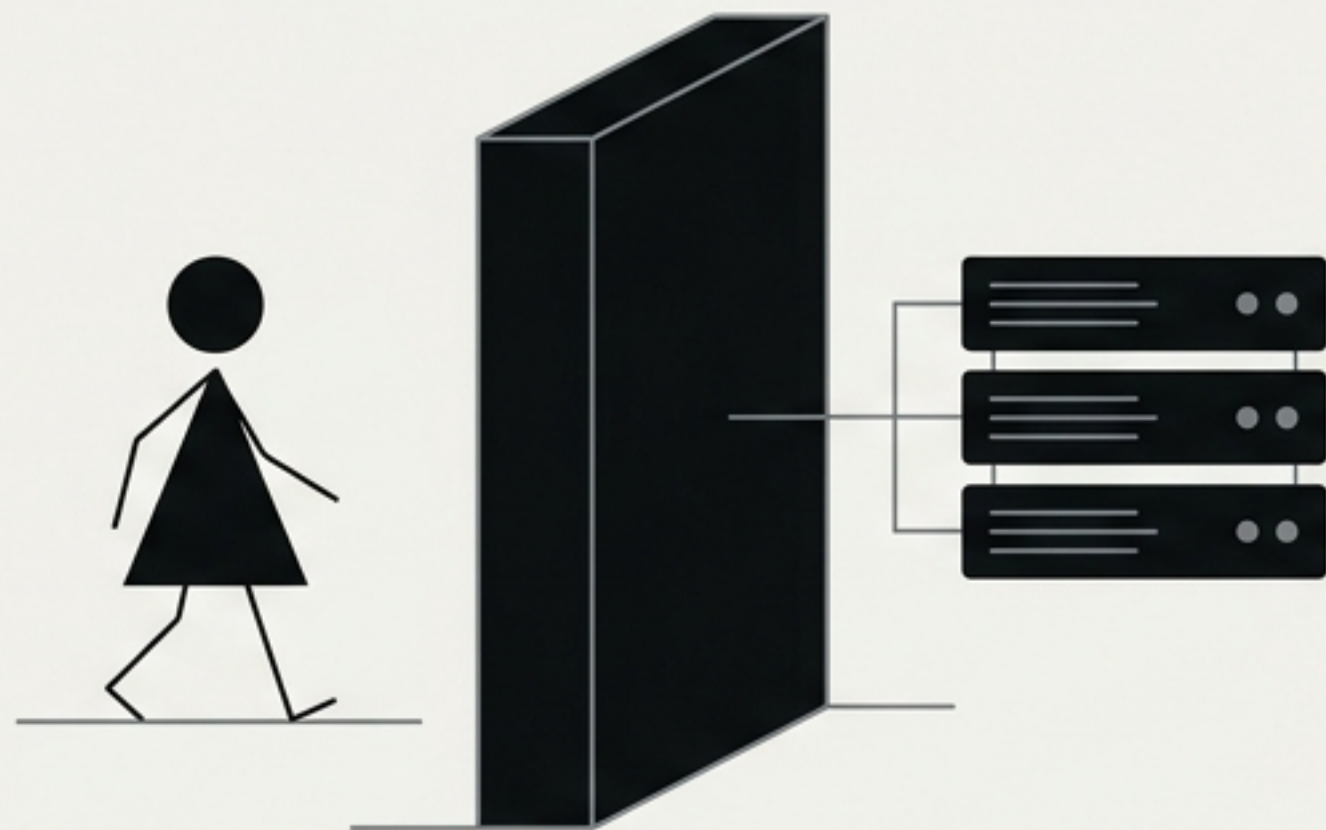




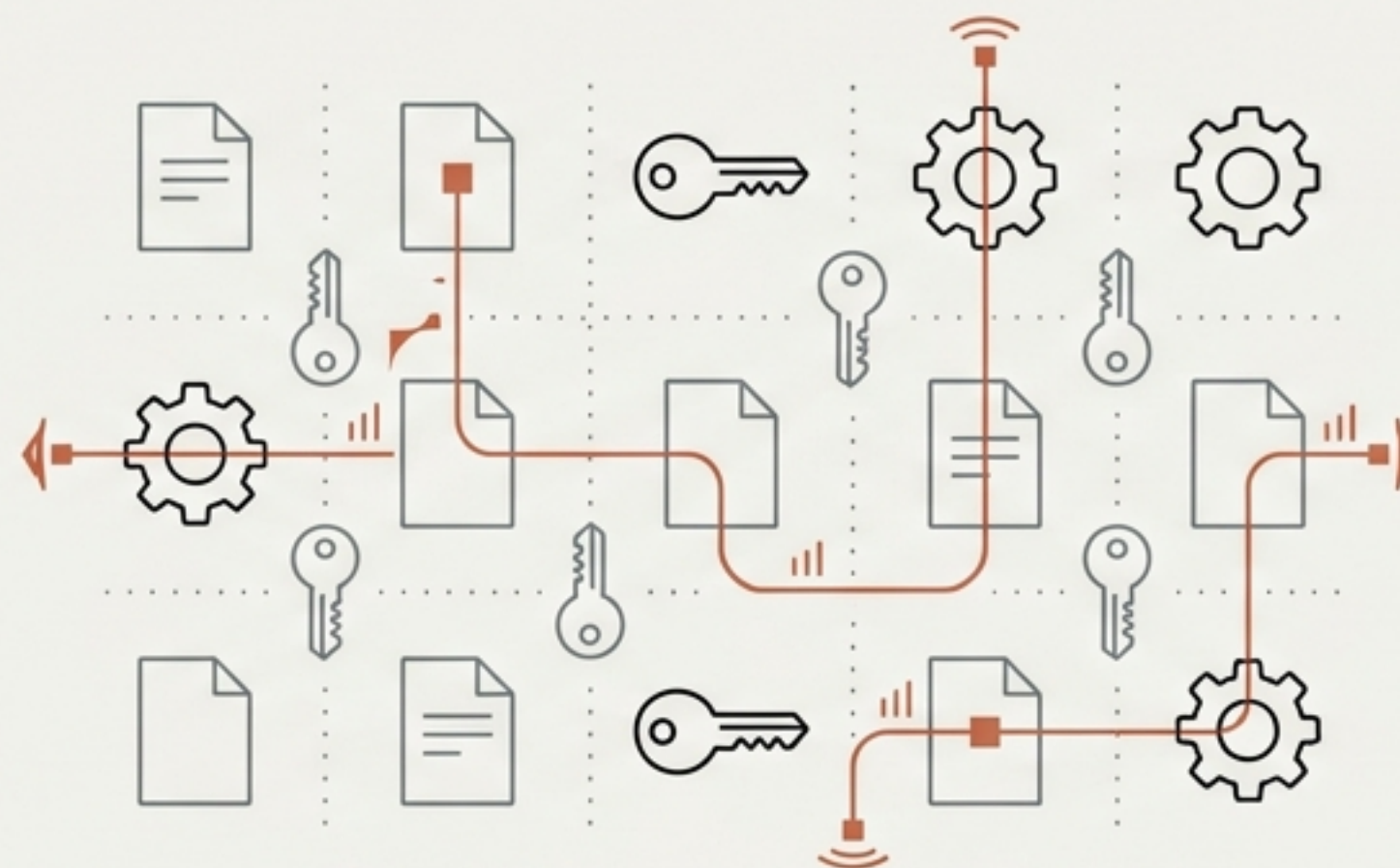
Defending the Intelligence Surface Area

Information Security in the Age of AI

The Core Security Assumption Has Broken



For decades, information security focused on protecting systems from people.



In the age of AI, organisations must increasingly protect themselves from intelligence.

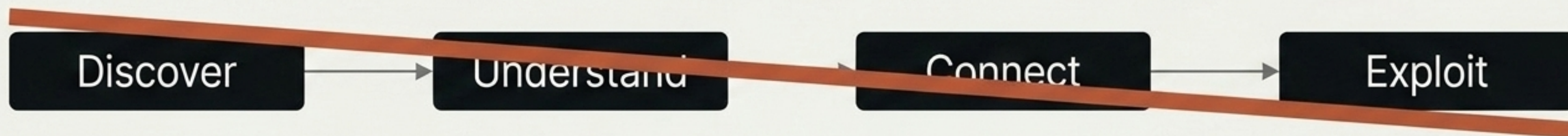
The objective is no longer simply reducing the attack surface.

The Paradigm Shift Matrix

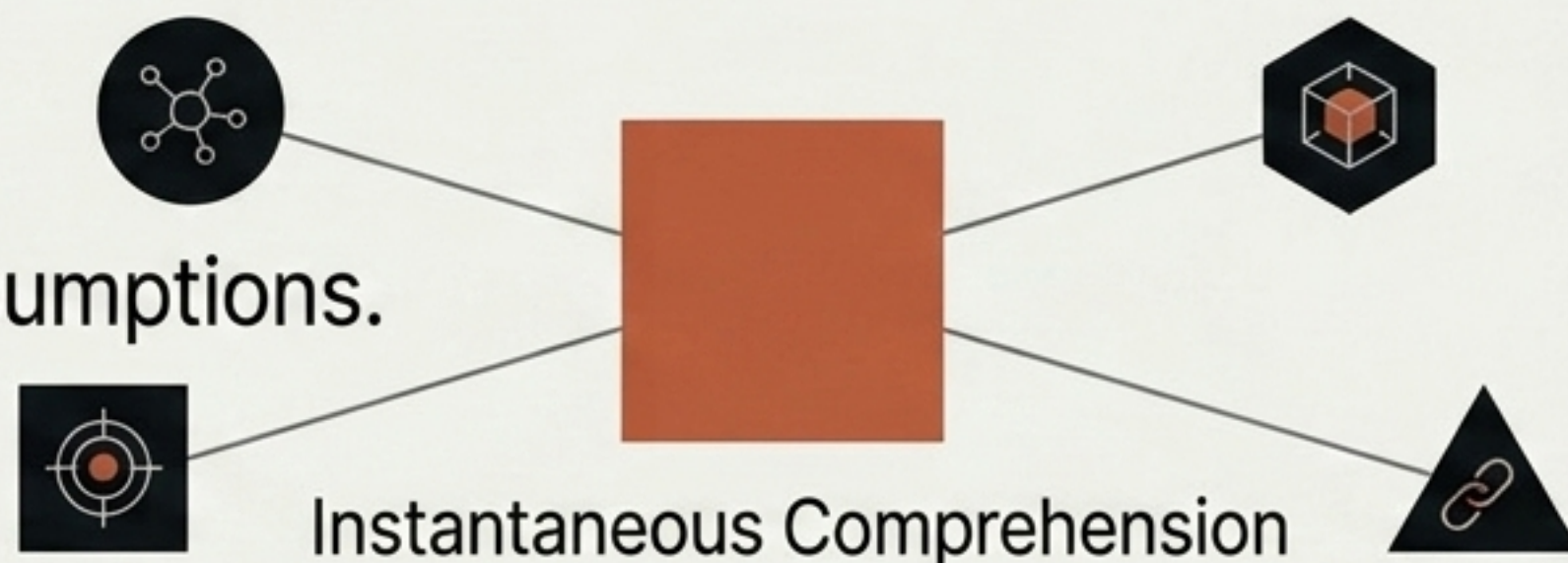
	Traditional Era	AI Era
Threat Actor	Humans	Autonomous Systems
Limiting Factor	Intelligence / Time	Available Context
Primary Strategy	Containment	Conduction & Reduction
Core Action	Adding Controls	Removing Exposure
Key Metric	CVEs / Pen Tests	Open Intelligence Surface Area (OISA)

AI Eliminates the Cost of Discovery

Traditional security assumes an attacker must manually discover systems, understand them, **connect** disparate info, and exploit weaknesses.

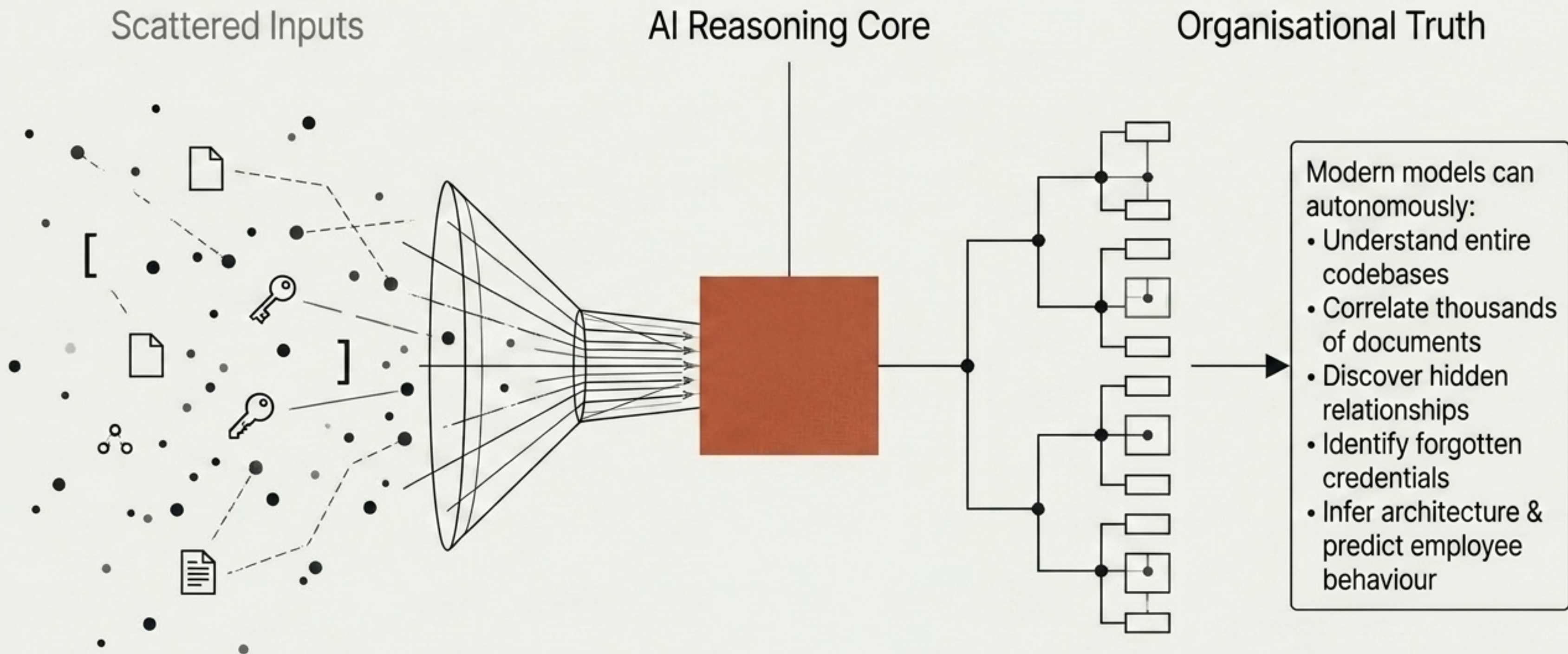


AI changes all four assumptions.



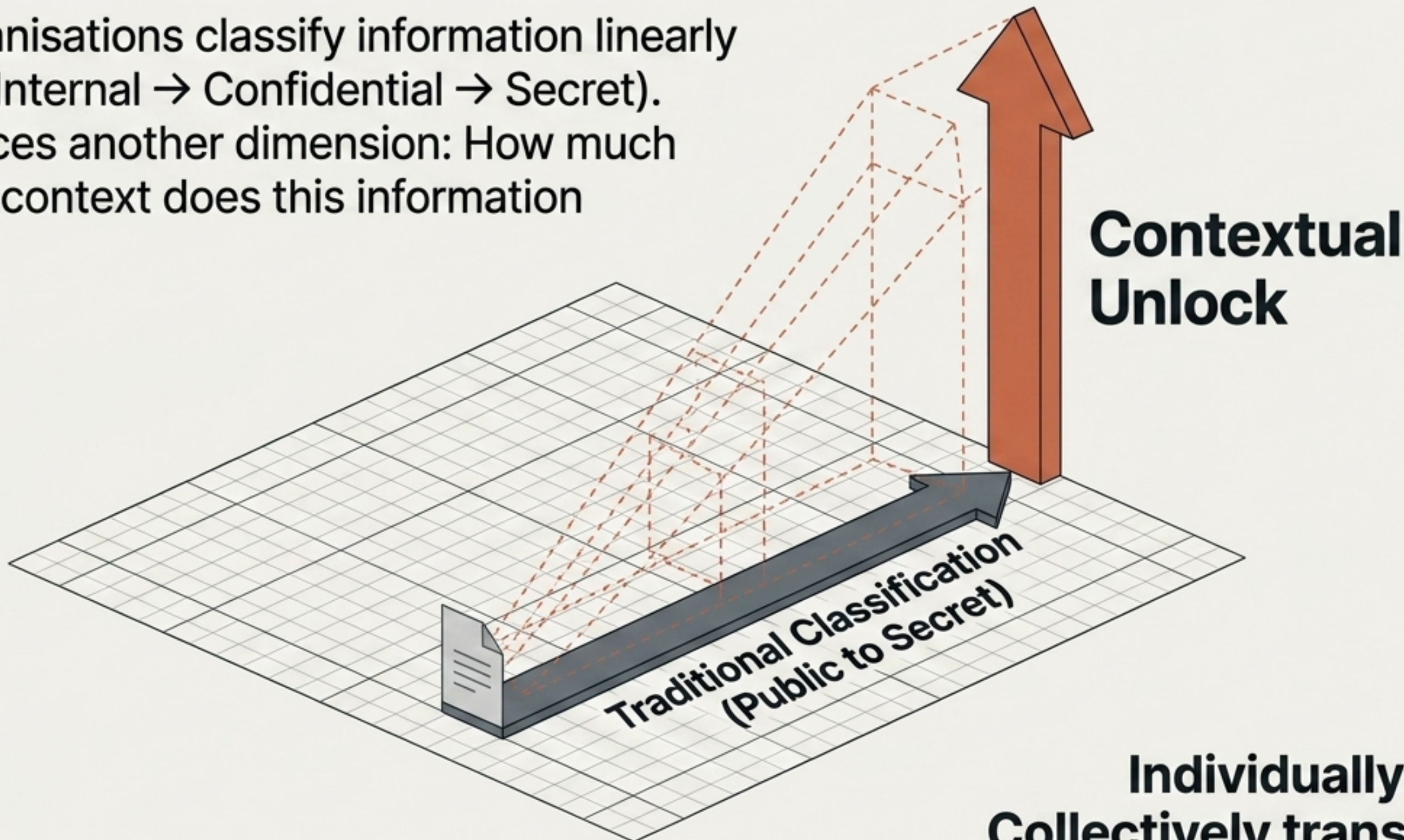
The limiting factor is no longer intelligence. It is available context.

The AI Reasoning Engine



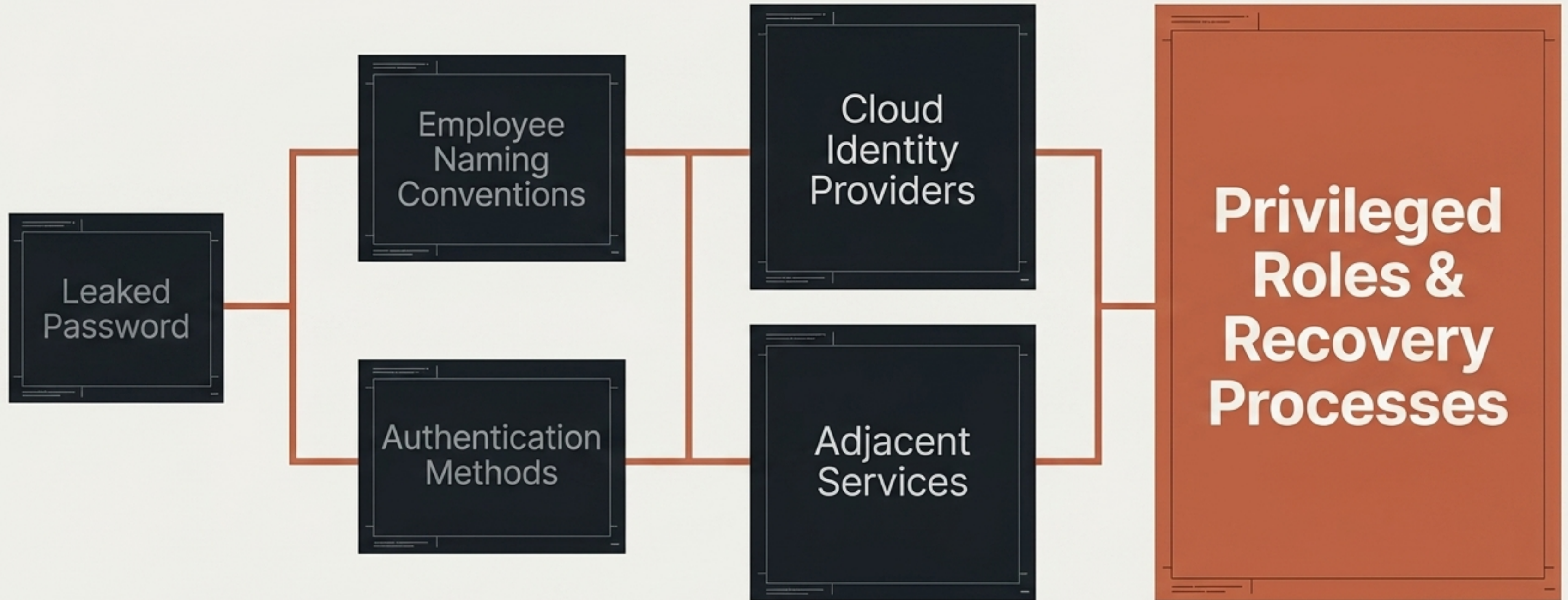
The Hidden Dimension of Data Value

Many organisations classify information linearly (Public → Internal → Confidential → Secret). AI introduces another dimension: How much additional context does this information unlock?



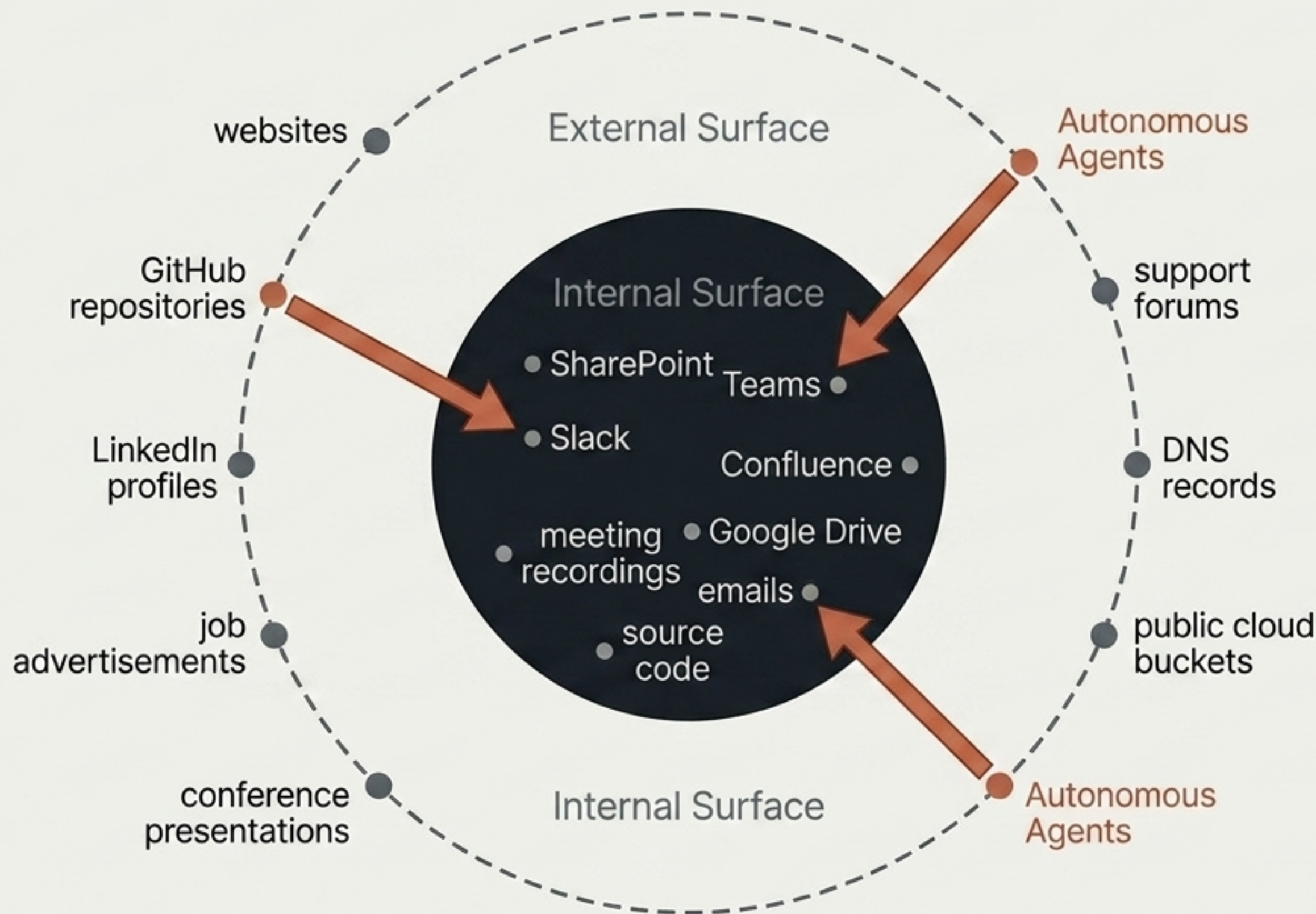
**Individually harmless.
Collectively transformative.**

How AI Makes Context Compound

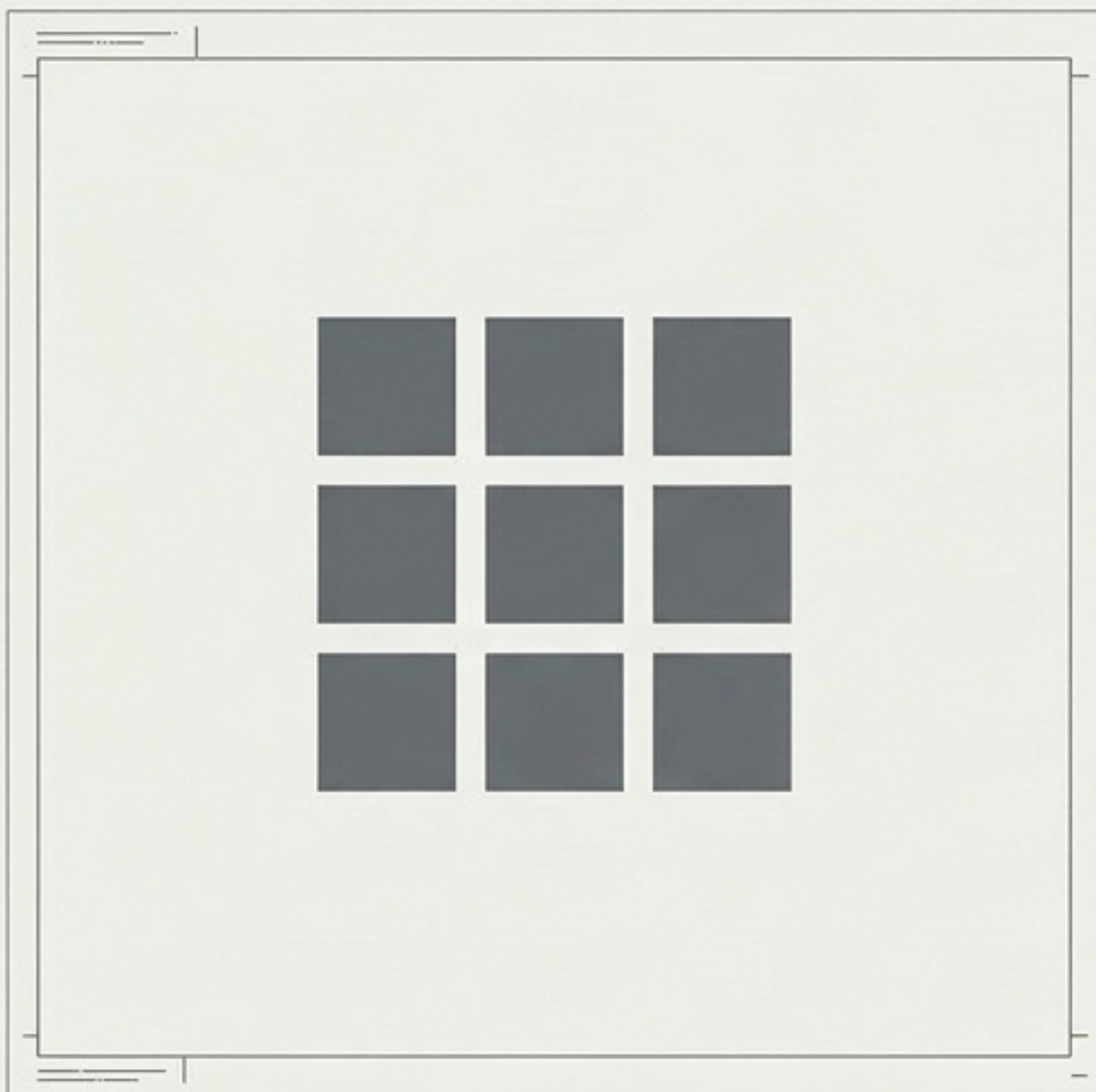


The value compounds because AI reasons across all available evidence simultaneously. One harmless dot unlocks the entire map.

The Open Intelligence Surface Area (OISA)



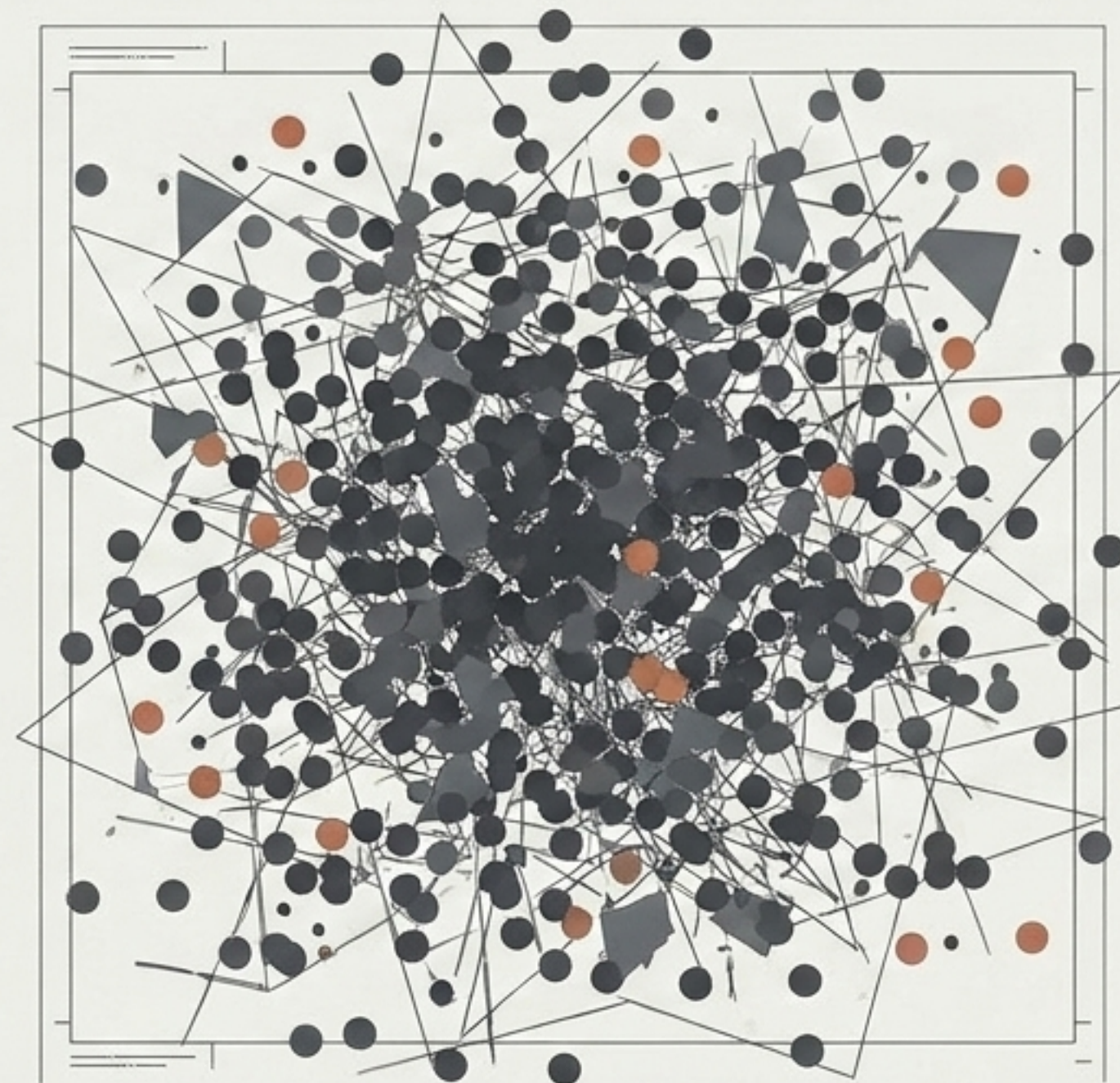
Informational Entropy



1. Storing information is cheap.

2. Retrieval is instantaneous.

3. Reasoning is effectively free.



Every unnecessary document, every forgotten spreadsheet, increases organisational entropy. High entropy equals high intelligence surface area.

The Equation of Vulnerability

~~Threat = Attacker Skill × Unpatched Vulnerabilities~~

$$\text{Threat} = \text{AI Reasoning Capacity (Approaching } \infty) \times \text{Exposed Context (Controllable)}$$

Because AI reasoning is effectively infinite and instantaneous, the only variable an organisation can still control is exposed context.

Security by Reduction

- | | |
|--------------------------------|----------------------------|
| • Do we need this document? | • Do we need this system? |
| • Do we need this integration? | • Do we need this account? |
| • Do we need this API? | • Do we need this data? |

Instead of continually adding security controls, remove unnecessary exposure. The safeset is the one that no longer exists.

Clearing Informational Debt

Just as organisations reduce technical debt, they must enforce an Information Diet to reduce informational debt.

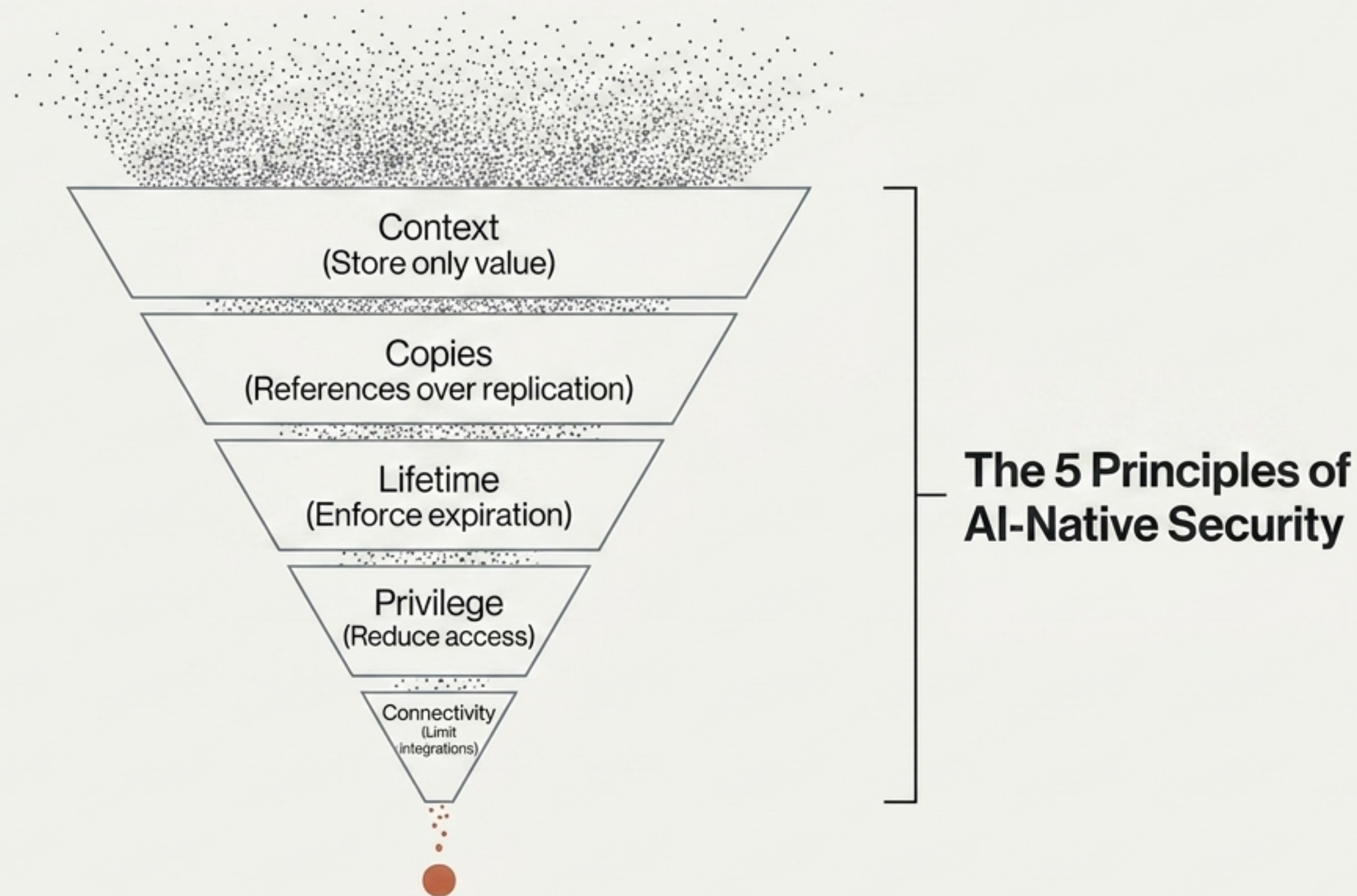


Regularly remove:

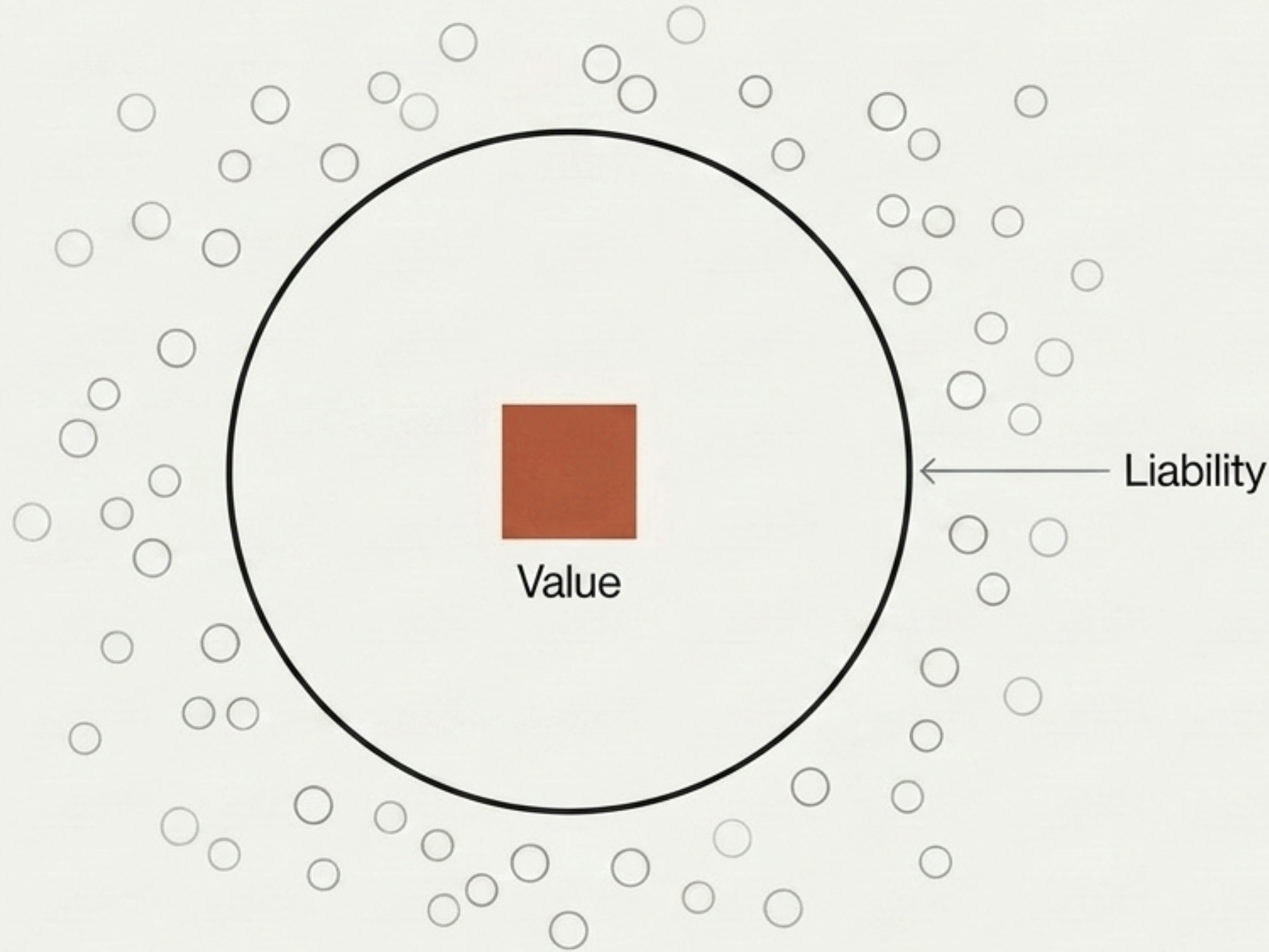
- Obsolete documents
- Unused repositories
- Inactive accounts
- Inactive accounts
- Duplicate knowledge bases
- Historical exports
- Unnecessary backups
- Abandoned cloud services

Less information = Less attack surface.

The Context Reduction Funnel



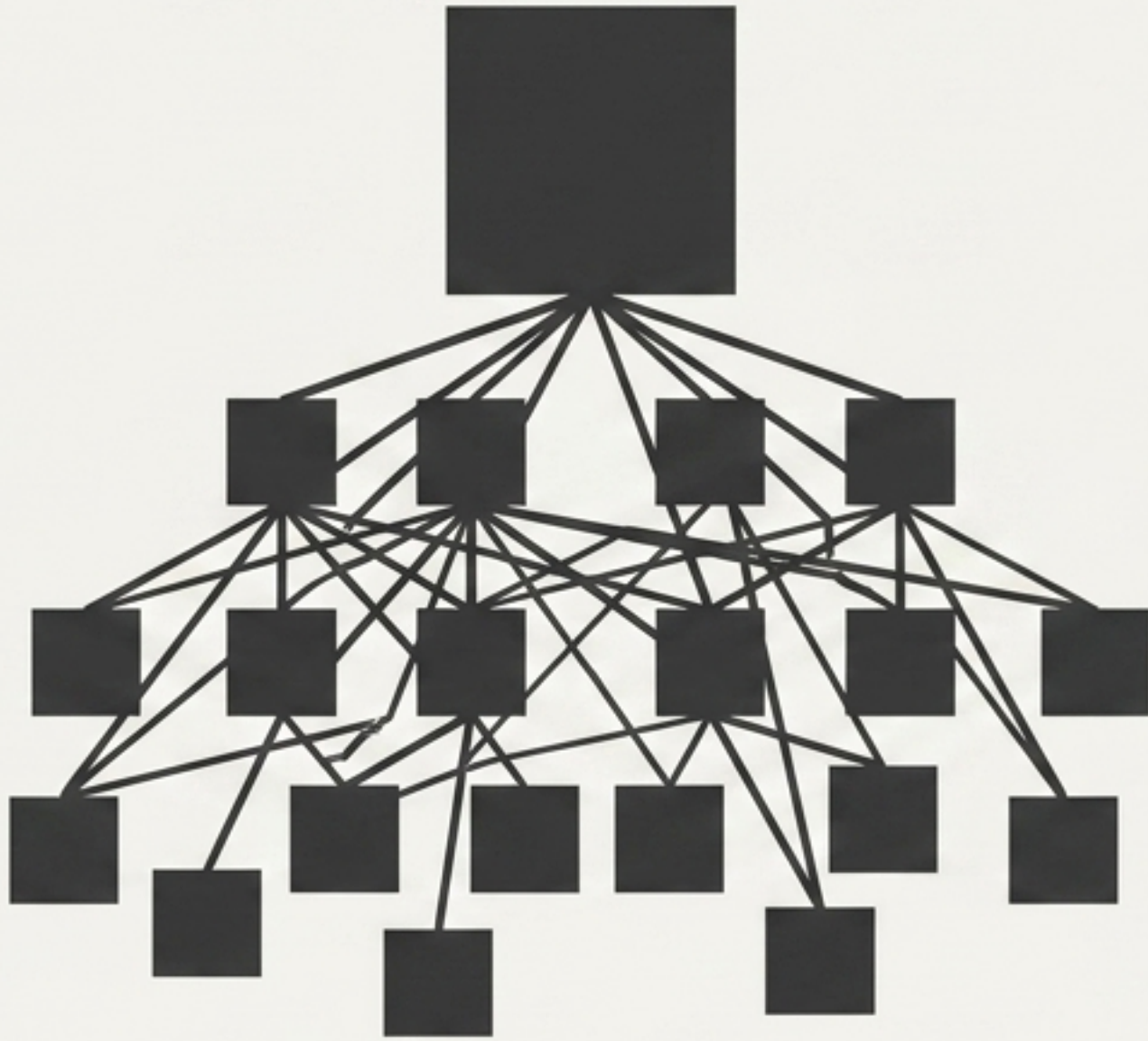
Principle 1: Minimise Context



**Store only what
creates value.
Everything else
becomes liability.**

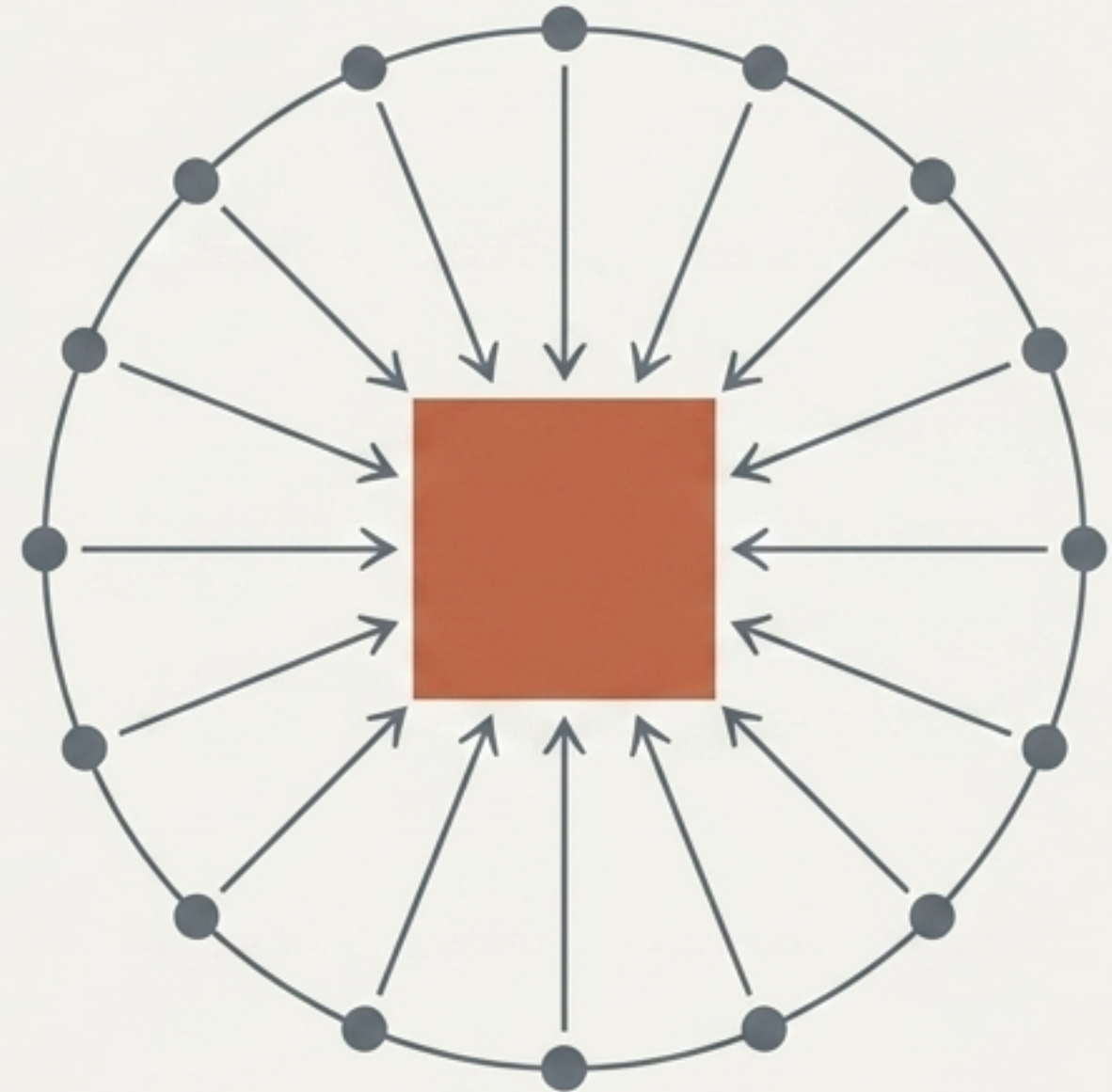
Principle 2: Minimise Copies

Replication



Every duplicate creates another exposure point.

Referencing



Prefer references over replication.

Principle 3: Minimise Lifetime

Permanent Retention (Infinite Exposure)



Expiration Enforced

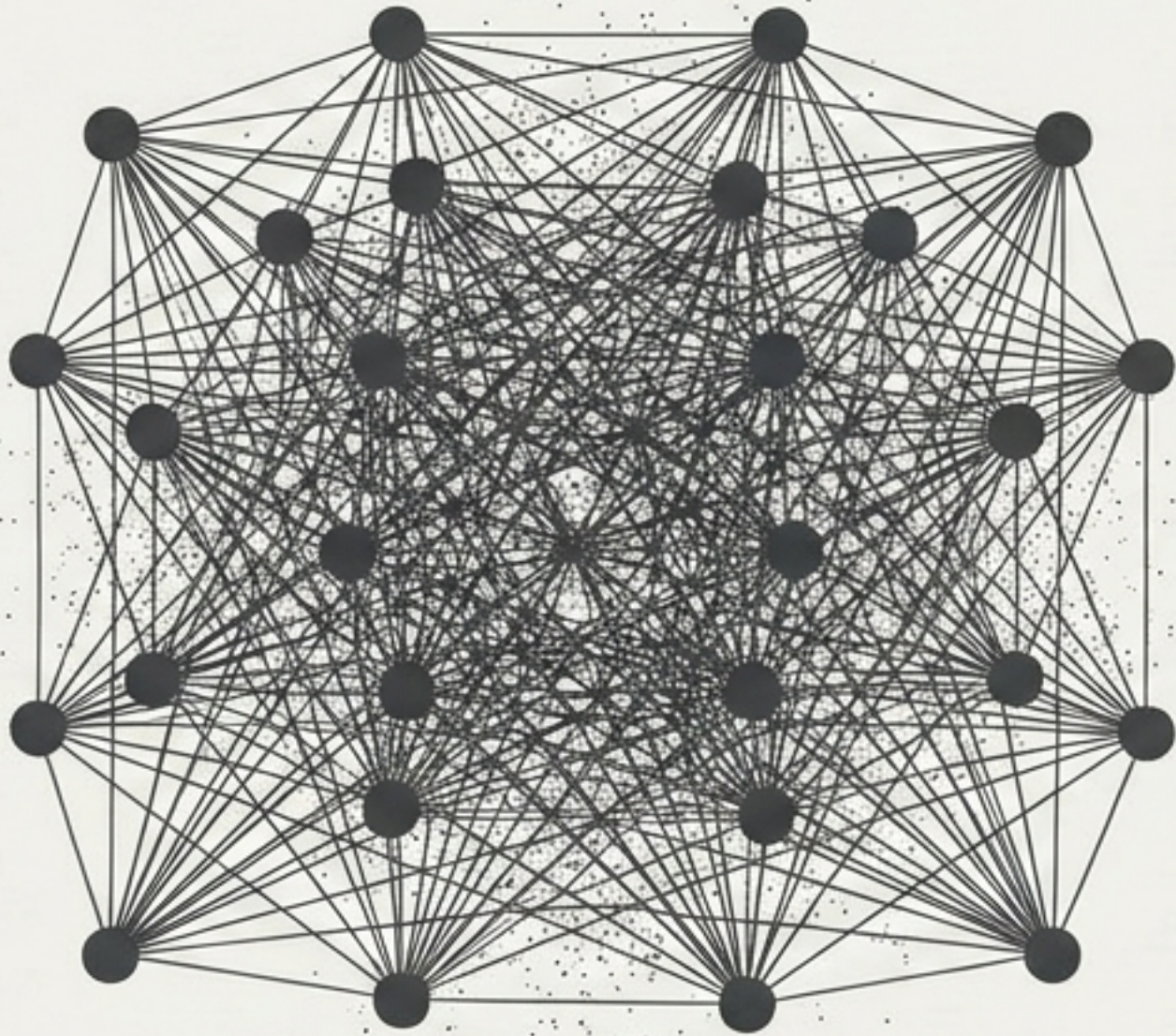


Information should expire.
Not everything deserves permanent retention.

Principle 4: Minimise Privilege

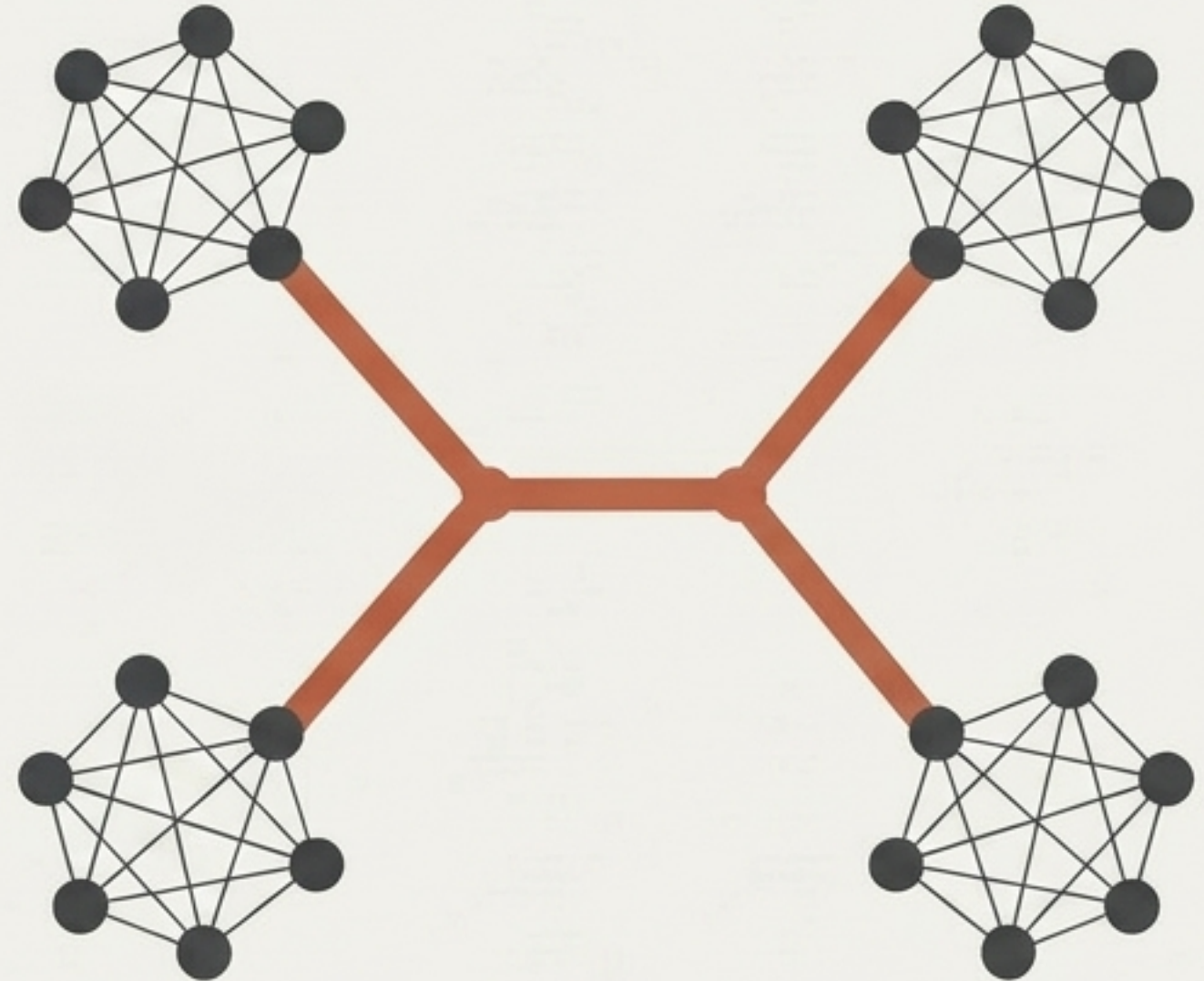
AI only needs one credential
to begin reasoning.
Reduce access continuously.

Principle 5: Minimise Connectivity



Exponential Intelligence Pathways

Every integration increases graph connectivity. Highly heavy policed pathways. Highly connected systems create exponentially more intelligence pathways.

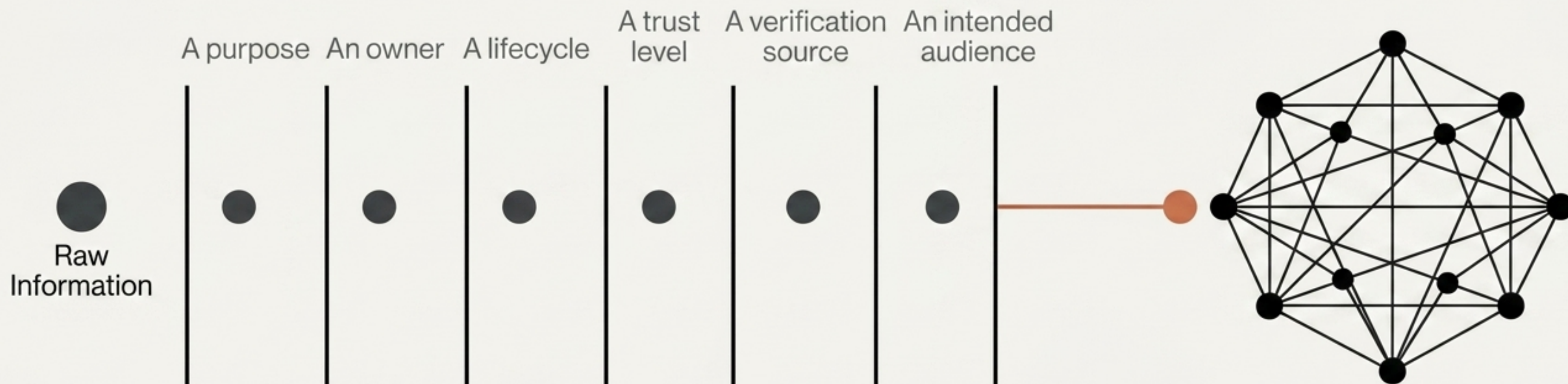


Governed Connections

The Future State: Verified Context Graphs (VCGs)

	Raw Unstructured Data	Verified Context Graphs (VCGs)
State	Millions of disconnected artefacts	Curated, verified relationships
Connections	AI inferred truth	Purpose-specific context
Leakage	High redundant leakage	Minimised / Purposeful
Governance	Difficult	Easy
Surface Area	Massive reasoning surface	Dramatically smaller

Conducting Rather Than Containing



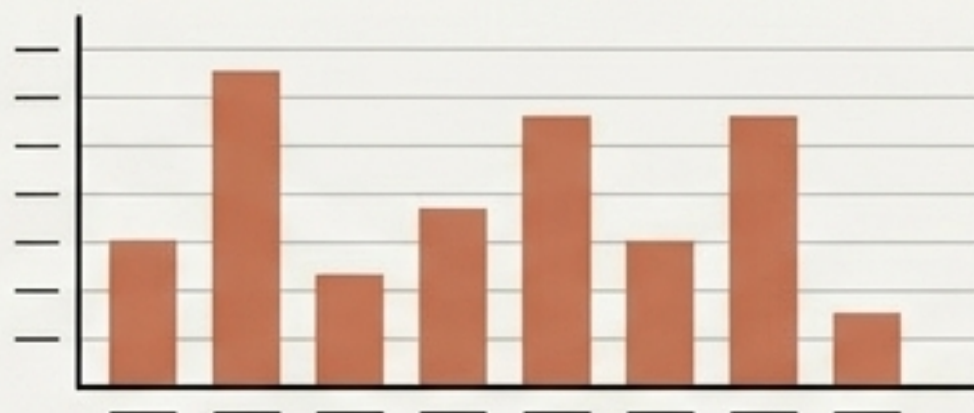
**Information without purpose becomes organisational noise.
Noise becomes attack surface.**

The OISA Dashboard: The New Security Metrics

Discoverable Information Volume



Disconnected Knowledge Stores



4,209

Duplicate Documents Exist



Dormant Identities Remaining



APIs Exposing Unnecessary Context



Unverified Organisational Knowledge



Reducing these metrics becomes as important as patching software.